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RESEARCH-ARTICLE

Decomposable and responsive power models for multicore processors using performance counters

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Decomposable and Responsive Power Models for Multicore Processors using Performance Counters

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ABSTRACT

Power modeling based on performance monitoring counters (PMCs) attracted the interest of researchers since it became a quick approach to understand and analyse power behavior on real systems. As a result, several power-aware policies use power models to guide their decisions and to trigger low-level mechanisms such as voltage and frequency scaling. Hence, the presence of power models that are informative, accurate and capable of detecting power phases is critical to increase the power-aware research chances and to improve the success of power-saving techniques based on them. In addition, the design of current processors has varied considerably with the inclusion of multiple cores with some resources shared on a single die. As a result, PMC-based power models warrant further investigation on current energy-efficient multicore processors.

In this paper, we present a methodology to produce decomposable PMC-based power models on current multicore architectures. Apart from being able to estimate the power consumption accurately, the models provide per component power consumption, supplying extra insights about power behavior. Moreover, we validate their *responsiveness* –the capacity to detect power phases–. Specifically, we produce a set of power models for an Intel® Core™ 2 Duo. We model one and two cores for a wide set of DVFS configurations. The models are empirically validated by using the `SPEC-cpu2006` benchmark suite and we compare them to other models built using existing approaches. Overall, we demonstrate that the proposed methodology produces more accurate and responsive power models. Concretely, our models show a [1.89-6]% error range and almost 100% accuracy in detecting phase variations above 0.5 watts.

Categories and Subject Descriptors

C.4 [**Performance of Systems**]: Modeling Techniques

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General Terms

Measurement, Experimentation

Keywords

Power estimation, Performance counters

1. INTRODUCTION

Energy, power density and power consumption have attracted the interest of researchers since they have become limiting factors in processor designs [22, 29]. Power density reduces the reliability and lifetime of processors [32] and limits their operating frequency [7]. Moreover, energy and power consumption are key factors in all market segments. For instance, they are important to extend the battery life of mobile devices and also, to reduce the need of power supply and energy bills of HPC data centers [8]. As a result, there are several techniques –working at different levels– addressing power related issues. All of them rely on accurate methods to gather information about power consumption. Specifically, for software-based solutions the need to estimate and predict power behavior has justified the research on power modeling methods.

The methods based on Performance Monitoring Counters (PMCs) have been shown to be a good solution to estimate power consumption. As a result, their applicability has been demonstrated on several fields such as power management and application profiling. PMC-based power models are used to perform live predictions of power behavior in order to guide power aware policies [20, 31, 4]. Moreover, they are also used in research for quickly exploring new approaches since they allow to profile real systems and full executions of applications, avoiding the need of performing long-time and limited simulations [3, 14, 19, 6, 18, 5]. In the end, they have been crucial in the process of addressing power issues.

The simpler approaches to produce PMC-based power models select the PMCs that are most correlated with power and apply a multiple linear regression to derive the model [3, 19, 6, 5]. This leads to accurate power models, but they do not allow to know how the power is consumed within the processor. In addition, these models are less acceptable to experts and layman because PMCs that should positively affect the final power consumption may have negative coefficients and vice versa. Figure 1 illustrates the importance of having a model capable of breaking down the power consumption. The figure shows two time slices of the `473.as-`

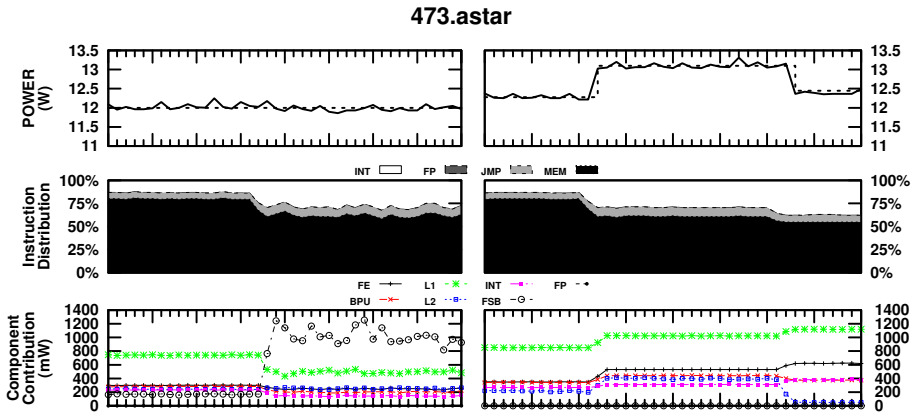


Figure 1: Power, instruction distribution and per component power breakdown for two time slices of the `473.astar` SPECcpu2006 benchmark.

`tar` application. During the first time slice on the left, the power consumption –at the top– remains constant. However, the distribution of executed instruction –at the center– and the per component power breakdown[1] –at the bottom– show variations. The Front Side Bus (`FSB`) contribution increases, the contributions of the other components decrease and the overall power does not change. In this case, just reading power measurements or using a model unable to breakdown the power consumption would not detect such variations. Specifically, in this example the data locality changed and data had to be brought from main memory. The time slice on the right shows that the changes of contribution of each component have not always the same direction as the changes of power. In this example, a model just based on the frontend (`FE`) component would fail to report the second change on power consumption because its contribution increases but the overall power consumption decreases. These two examples demonstrate the advantages and the necessity of tracking as many components as possible in order to model correctly the power behavior.

More complex PMC-based modeling methods solve these issues by splitting the model in microarchitectural components and then use heuristics –such as floorplan information– and manual tuning (not systematic) to calibrate the model. These last techniques have been successfully applied to previous generation processors such as Pentium IV [14]. Since these processors were not designed taking into account power consumption as a main design constraint, they lack or do not use extensively several power related mechanisms [10, 17] that are common in today's processors. Moreover, the design of current processors has varied considerably with the inclusion of multiple cores with some resources shared on a single die. As a result, these decomposable power models warrant further investigation on current energy-efficient multicore processors. The proposal in this paper performs such investigation on such architectures and presents a new systematic methodology that reduces the complexity of generating decomposable power models for current multicore architectures, avoiding the need of using heuristics or manual tuning. Our proposal is a tradeoff between the component breakdown granularity of the model and the complexity of producing it. As a result, this approach addresses the problem of the lack of decomposable power models for current architectures.

[1]The power components are mapped to the microarchitectural components in the following manner. FE: frontend, INT: integer units, FP: floating point units, BPU: branch prediction unit, L1: L1 cache, L2: L2 cache and FSB: front side bus and main memory.

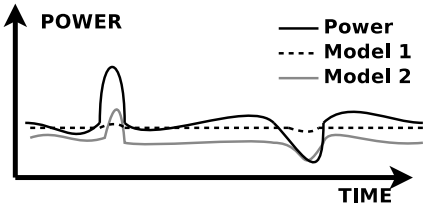


Figure 2: Power model examples.

Besides, the validation of power models have been only carried out via calculating the accuracy of their predictions. However, most of the power aware solutions that are built on top of the models, trigger their actions when they detect some variation in power consumption, i.e. a change of power phase. Therefore, it is important to study the capacity of the models to detect power phase changes because it affects the final success of the solutions built on top of them. Concretely, it must be validated that models detect correctly the magnitude and the instant of phase variations. Figure 2 illustrates the importance of this problem. The two models shown have low average error in their power predictions. In fact, Model 1 on average is more accurate than Model 2. In this case,a validation based only on accuracy would say that the Model 1 is more suitable than the Model 2. However, the Model 1 would not predict any change in power behavior and therefore, any policy built on top of it would miss optimization opportunities. The capacity of the Model 2 to guide power policies is better since it reacts in a similar fashion to power. We call this property the *responsiveness* of the model. We conclude that the overall usefulness and applicability of power models rely on their accuracy and also on their responsiveness. Consequently, both metrics should be evaluated in order to validate power models.

In summary, the properties that power models must have in order to improve their applicability are: accuracy, responsiveness and decomposability. In addition, the methodology

to produce them should be applicable on new energy-efficient multicore architectures under different power configurations –i.e. different frequency configurations–. The main contributions of this paper are:

- A new systematic methodology for producing PMC-based power models on multicore architectures. The methodology ensures the decomposability, the accuracy and the responsiveness of the models generated.
- We evaluate and validate the *responsiveness* of the models based on power phase detection accuracy.
- A case study is presented for an Intel® Core™ 2 Duo, a high performance processor designed with power efficiency as a main design constraint [12]. Single and multiple core models are presented for different DVFS configurations. An empirical validation of the produced models is performed using long executions of the `SPECcpu2006` benchmarks. The accuracy validation shows average errors between [1.89-6]%. The responsiveness validation when operating at maximum frequency shows an overall 83.31% accuracy on phase change predictions and near 100% accuracy for changes bigger than 0.5 watts. As a result, we show that the model produced is sensible to variations as small as 0.25 watts, which only represent a 2.24% variation of the average power consumption of the `SPECcpu2006` suite.
- A power phase characterization of the `SPECcpu2006` benchmark suite is presented.

The rest of the paper is organized as follows. Section 2 describes our modeling methodology including the power component definition, the microbenchmark design, the experimental infrastructure and the method to produce the power models. The validation of the power models is presented in Section 3, including a responsiveness analysis. In Section 4 we compare and contrast related work. Finally, Section 5 concludes by summarizing our results.

2. METHODOLOGY

In this section, we describe our methodology for producing power models. The discussion is guided through the presentation of an example for a particular architecture, an Intel® Core™ 2 Duo [12]. The final outcome of this section is a set of power models with three main characteristics. First, the models are able to breakdown the power consumption among several architectural components. Second, the models have low average errors in their predictions. And last, the models are responsive in front of power variations. These last two characteristics are validated in Section 3 and the first one is ensured by the methodology.

In short, our methodology to produce decomposable power models follows the common modeling steps. First, we define the model inputs, which in our case are the power component activity ratios. Second, we define the training data which in our case is generated using microbenchmarks. Third, we collect the required data in order to train and validate the model respectively. And fourth, the model is built using our method. The main differences of our methodology over previous works are three: the basic rules to define power components; the design of the microbenchmarks, which isolate

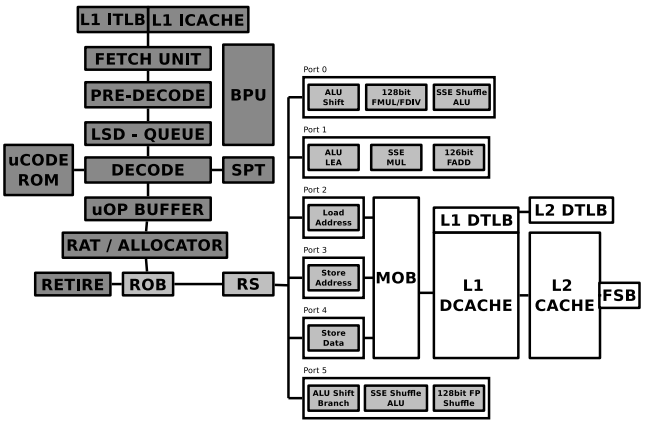


Figure 3: Single core view of Intel® Core™ 2 Duo microarchitectural components.

and decouple power component activities allowing the generation of decomposable power models; and finally, the model generation methodology, which profits from the specifically generated training data to built accurate, responsive and decomposable power models.

It should be noted that even though the example presented is for a particular architecture, the methodology can be applicable to other architectures. Particularly, one should redefine and redesign the power components and the microbenchmarks in order to fulfill the requirements of our method. The rest of the section presents in detail the aforementioned steps and justifies their rationale.

2.1 Defining power components

The inputs of our model are component activity ratios, which have been shown to be good proxies for power estimation [3, 14]. As a result, the first step towards the final power model is to define the power components and their associated activity ratio formulas.

In order to have a detailed power model, we need to reach a component definition with as many power components that represent a single microarchitectural component as possible. However, there are two main restrictions to this direct-mapping between microarchitectural components and power components. First, some microarchitectural components are tightly related in the sense that even if they are monitored by a different set of PMCs, they expose very similar levels of activity. For such components it is not possible to discern directly their power contribution because their activity cannot be isolated or decoupled from the activity of other microarchitectural components. And second, there are also microarchitectural components that do not have any PMC that reports their activity. Therefore, due to these restrictions, in our approach a power component can represent one or a set of microarchitectural components. This approach is a reasonable tradeoff between the undecomposability of simpler approaches [31] and the complexity of the microarchitectural-level decomposable ones [14].

As a case study, we explain the power component definition for an Intel® Core™ 2 Duo. We have identified more than 25 microarchitectural components on this architecture. Figure 3 shows an schematic view of them classified into three categories in order to facilitate the explanation: the

Table 1: Power components defined for an Intel® Core™ 2 Duo in conjunction with their PMCs-based activity formulas and microarchitectural components that represent.

Power Component	Activity formula	Modeled components
FE	UOPS_RETIRED:ANY/ CPU_CLK_UNHALTED:CORE_P	L1_ITLB, L1_ICACHE, FETCH_UNIT, PREDECODE, LSD, DECODE, uCODE ROM, SPT, uOP BUFFER, RAT, ROB, RETIRE
INT	(RS_UOPS_DISPATCHED_CYCLES:PORT_0 + RS_UOPS_DISPATCHED_CYCLES:PORT_1 + RS_UOPS_DISPATCHED_CYCLES:PORT_5 - FP_COMP_OPS_EXE - SIMD_UOPS_EXEC - BR_INST_RETIRED:ANY)/ CPU_CLK_UNHALTED:CORE_P	Integer arithmetic units
FP	FP_COMP_OPS_EXE/ CPU_CLK_UNHALTED:CORE_P	Floating point arithmetic units
SIMD	SIMD_UOPS_EXEC/ CPU_CLK_UNHALTED:CORE_P	SIMD arithmetic units
BPU	BR_INST_DECODED/ CPU_CLK_UNHALTED:CORE_P	BPU and branch execution
L1	L1D_ALL_REF/ CPU_CLK_UNHALTED:CORE_P	LD/ST execution, MOB, L1, L1 DTLB, L2 DTLB
L2	L2_RQSTS:BOTH_CORES:ANY:MESI/ CPU_CLK_UNHALTED:CORE_P	L2
FSB	BUS_DRDY_CLOCKS:ALL_AGENTS/ CPU_CLK_UNHALTED:BUS	FSB and memory

in-order engine (dark gray), the memory subsystem (white) and the out-of-order engine (light gray).

The in-order engine includes some microarchitectural components that do not have PMCs that report their activity. Moreover, the activity of each of these components cannot be isolated since they are in the in-order part of the pipeline, i.e. activity in the `FETCH` unit, means activity in the `PREDECODE` unit. Therefore, we group the whole in-order engine components –excepting the branch prediction unit (`BPU`)– as one power component, namely, the frontend (`FE`) of the processor. The decision of defining the `BPU` power component separately is based in the fact that it includes several predictors that can consume between 10% and 40% of the whole processor power budged [23].

For the memory subsystem, it is not possible to generate activity in one memory level without causing activity in the previous one due to its stacked configuration. However, it is possible to decouple their activity. As a result, the whole memory subsystem is divided in three power components: the `L1` cache, the `L2` cache and the Front Side Bus (`FSB`), which also represents the main memory.

The parts of the architecture in the out-of-order engine define the `INT`, `FP` and `SIMD` power components. At the end, eight power components are defined. Table 1 summarizes them in conjunction with their related microarchitectural components and PMC-based activity formulas.

2.2 Designing microbenchmarks

We need empirical data in order to train our power model. For that purpose, a set of microbenchmarks must be designed. This is an important step in the methodology because not considering a significant set of possible scenarios incurs in inaccuracy of the final model [33]. We do not use real applications to train the model due to two reasons. First, we want a model suitable for any workload; therefore, the training set should be workload independent. And second, in order to be able to derive the contribution of each power component to the total power consumption it is mandatory to be able to isolate or decouple their activity. This is required in order to apply multiple regression techniques without collinearity problems [24].

The microbenchmark-based approach is always a source of inaccuracy since it is not possible to cover all scenarios of power consumption and activity. For example, for exercising the `INT` units we can use different instructions that will consume differently. Similarly, different inputs produce different power consumption. In conclusion, the microbenchmark suite should cover enough variety of power behavior in order to minimize all these sources of inaccuracy.

For our case study, we have designed 97 microbenchmarks. Their structure is an infinite loop containing a specific sequence of assembly instructions with distinct dependency chains in order to explore different activity ratios. However, we had to face the problem of the inherent correlation between the `FE` component and the rest of components. We solved it by introducing diverse combinations of `fxch`[2] instructions. The instruction sequence within the infinite loop is long enough to minimize the effect of the branch instruction at the end of it. Table 2 summarizes the activity ratio ranges that are explored for each power component. For example, for the `INT` component we have designed 13 benchmarks with `INT` activity ratios ranging from 1 to 3 and `FE` activity ratios from 1 to 3.45. Notice that we are able to have a `FE` activity of 3.45 uops/cycle with only 3 uops/cycle on the `INT` component, while the rest of components unused. This decoupling of activity is the effect of the `fxch` instruction and it is mandatory to be able to form a decomposable power model afterwards. Next section describes the experimental framework and the data gathering process used to obtain the power consumption.

2.3 Collecting Data

All the experiments are carried out on a workstation that features an Intel® Core™ 2 Duo T9400 processor [12] and two modules of 2GB of memory. The embedded controller firmware of the platform provides information about the power source and fulfills the SMAPI specification [30]. This specification defines an interface to obtain power consumption measurements with a guaranteed granularity and accuracy. As a result, we are able to gather power measurements in a granularity of milliwatts with a maximum error of 2%.

The platform runs a Linux kernel 2.6.28 with the required patches to allow PMC readings. The `tp_smapi` [1] module is loaded in order to be able to gather the power information. This module creates some entries in the `/sys` filesystem that provide information such as the power consumption. We use a modified version of `pfmon` [2] to access to the PMCs and power consumption simultaneously. All measurements are obtained running the experiments in standalone mode to minimize interferences. We switch off all sources of power consumption –i.e. the display– that we do not want to include in the model. The components of the platform where it is not possible to switch them off are configured at constant operation mode. For example, we find that the fan of the processor introduces up to 0.5W variations on power consumption. Hence, we configure it to always operate at full-speed. Under these conditions, we track the power consumption during five minutes before each experiment and we find that the baseline power consumption of the idle system

[2]As far as we know, the *fxch* is the only instruction that does not dispatch any operation to the out-of-order engine. It only performs register renaming, therefore it allows to decouple the activity on the `FE` and the components in the out-of-order engine.

Table 2: Microbenchmark characteristics. Activity ratios and power consumption ranges.

Microbench set	#	FE Activity	INT Activity	FP Activity	SIMD Activity	BPU Activity	L1 Activity	L2 Activity	FSB Activity	Power Range
FE	1	1	0	0	0	0	0	0	0	5587mW
INT	13	1-3.45	1-3	0	0	0	0	0	0	5792mW-8114mW
FP	9	0.2-1.98	0	0.2-1	0	0	0	0	0	5087mW-6887mW
SIMD	12	1.85-3.29	0	0	0.99-2.63	0	0	0	0	6966mW-9274mW
BPU	5	0.42-1.14	0-0.42	0	0	0.46-1	0	0	0	5761mW-8521mW
L1	16	1-2.97	0	0	0	0	0.66-2	0	0	6571mW-8160mW
L2	12	0.12-0.42	0	0	0	0	0.11-0.22	0.11-0.21	0	8600mW-10112mW
FSB	18	0.02-0.14	0	0	0	0	0.02-0.04	0.02-0.04	0.58-0.71	10976mW-12205mW
MIX	11	1.63-3.95	0-1	0-0.8	0-1.97	0-0.34	0-1.97	0-0.07	0-0.34	7318mW-11298mW
Total	97	0.02-3.95	0-3	0-1	0-2.63	0-1	0-2	0-0.21	0-0.71	5087mW-12205mW

is 9.8W. The fact that we find a fairly constant idle power consumption for all the experiments demonstrates that we nullify possible interferences. For the purpose of the model generation, we assume that baseline, 9.8W, to be the power consumption of the platform except the processor and the memory. This assumption and the accuracy ensured by the SMART specification avoids the need of more complicated measuring techniques [31].

The performance monitoring unit of the processor is not able to track all the selected PMCs simultaneously. Therefore, we group them in sets that can be sampled at the same time. We program `pfmon` to switch PMC sets every 10ms and print the PMCs values and power measurements every 2 seconds. Although it is widely accepted that the PMC sampling does not introduce unacceptable inaccuracies [16, 14], we measure its effect. The sampling introduces interferences of less than 150mW.

To generate the input data for training the power model we run every microbenchmark during three minutes with only one core enabled, collecting a trace of 90 samples. We validate that each microbenchmark stress the component it is designed to do so by applying the formulas in Table 1. Moreover, we check that during the entire trace of each microbenchmark the PMC activities and power consumption remain nearly constant. We use the average over the 90 samples so that the possible errors and noise are minimized as much as possible [33]. Similarly, we collect the data for 26 applications of the `SPECcpu2006` [13] benchmark suite. In this case, every benchmark runs for 30 minutes, or less if it ends earlier. This data is going to be used to validate the power models.

All gathered data allows to explain the difficulties to direct relate the processor activity and the actual power consumption. In order to show the type of gathered data, Figure 4 depicts a time slice of the activity evolution of all components for three applications of the `SPECcpu2006` benchmark suite. We accompany this data with actual power measurements and the instruction distribution between load, store, integer, floating point and branch instructions which can be computed from the activity ratios. In general, we observe that power variations might have very different causes related to specific trends on every component. We do not detail them to avoid overloading the explanation. However, from that observation we conclude that power aware policies built on top of decomposable power models have more opportunities because they can discern the causes and components involved in power variations. Moreover, these power aware policies can also take advantage of the extra information provided by the model inputs –i.e. component performance or instruction distribution–.

2.4 Modeling power

2.4.1 Single Core modeling

We model the power consumption using a multiple linear equation with seven input variables, one for each power component defined. The `SIMD` component is not used because the `SPECcpu2006` suite do not present any activity in such component. Therefore, the total power consumption is expressed as:

$$P_{total} = (\sum_{i=1}^{i=comps} AR_i \times P_i) + P_{static} \quad (1)$$

where P_i is the weight of component i that we need to solve, and AR_i is its activity ratio. The $AR_i \times P_i$ represents the dynamic power consumption of component i, and P_{static} represents the overall static power consumption of all components. This approach is commonly used for generating PMC based power models [14, 33, 31]. However, instead of applying directly linear regression techniques [11, 24], we derive the marginal effect of each component to the overall power consumption. Firstly, we compute the P_i of each component –except the `FE`– separately using a multiple linear regression with the component related microbenchmark set as input. For example, we computed P_{INT} using only the `INT` microbenchmark set, which only stresses the `INT` component. The components which activity is not completely isolated –i.e. `L2` and `FSB`– are calculated incrementally. For example, before deriving P_{L2} from the `L2` microbenchmark subset, we derive P_{L1} and use that value for the P_{L2} estimation. The rationale that justifies this methodology is that the collinearity that may exist between components that are not stressed together is cancelled. Therefore, we get better estimates about what is the contribution of each component. The goodness of the estimates also relies on the collinearity between the exercised component and the `FE`, which we minimize with the specifically designed microbenchmarks. Finally, we use again a multiple linear regression using as input the `MIX` microbenchmark subset to derive P_{static} and P_{FE}. We use that set because it represents the common activity scenario, when there is activity on all processor components, canceling the known effect of under-estimate their contribution [33]. Moreover, this process is automatized using an R script [28] in order to be applicable systematically. This is possible because no manual tuning is required, demonstrating that with an appropriate set of microbenchmarks it is possible to generate decomposable and accurate power models.

Table 3 summarizes the models generated. The first four columns indicate the model name, the methodology used to

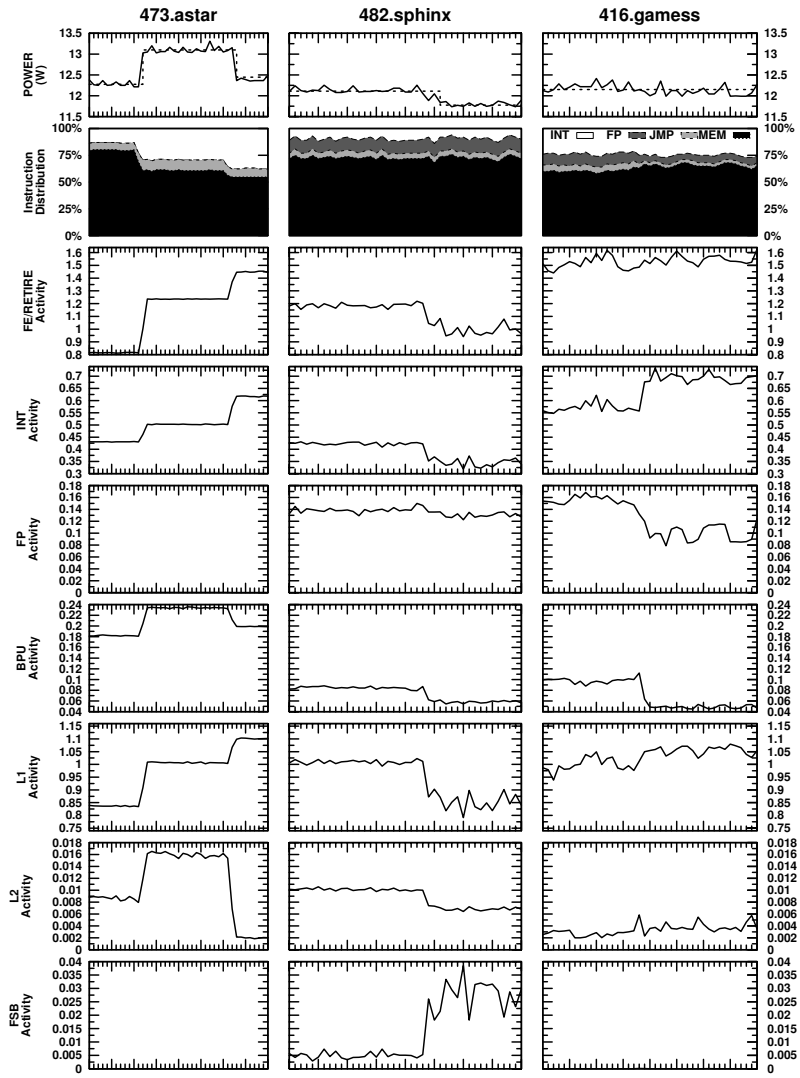


Figure 4: Power consumption, instruction distribution and component activity ratios during three time slices of the `473.astar`, `482.sphinx` and `416.gamess` `SPECcpu2006` benchmarks.

produce the model, the training dataset and the frequency used to generate it respectively. Using our methodology (`inc.` in Table 3), we have generated one model for each frequency available in our experimental platform. We refer to our models as `MICRO` models and can be found at the top eleven rows in Table 3.

Analysing in detail the `MICRO` models in Table 3, one can see that the weight of all the components increases exponentially with frequency. An exception is the `FSB` component which remain fairly constant on two values. This is because the frequency of the memory bus does not change with the frequency of the processor. Actually, it is 133GHz for core frequencies below 1.6GHz and 266GHz otherwise. The fact that the power weights have a exponential relation –like power–, provides an evidence about the effectiveness of the proposed methodology to model power behavior.

We have also produced other models using already known techniques to able to compare them, in Section 3, against our `MICRO` models. To generate the `MICRO-LIN` model, we have applied a multiple linear regression using as input all the microbenchmarks. Moreover, we produced a set of over-trained models taking an incremental number of input variables. The `FE` model only uses as input the `FE` activity and the `+FSB` model uses the `FE` activity plus the `FSB` activity. These models are also generated applying multiple linear regressions but using as input the `SPECcpu2006` data. Notice that these last models do not fulfill our objectives because they do not track all the components, the component weights are not acceptable to decompose power –i.e. negative weights– or they are not workload independent.

2.4.2 Multiple Core modeling

We use an accumulative approach for modeling multiple cores assuming that each core behaves equally. This assumption have been already done in similar works [31]. Hence, we apply the single core model to each core in the architecture. Therefore, we express the total power consumption as:

$$P_{total} = \sum_{j=1}^{j=cores} \left(\left(\sum_{i=1}^{i=comps} AR_{ij} \times P_i \right) + P_{static} \right) \quad (2)$$

Table 3: Models generated for an Intel® Core™ 2 Duo.

Model	Method	training set	Frequency	P_{FE}	P_{INT}	P_{FP}	P_{BPU}	P_{L1}	P_{L2}	P_{FSB}	P_{STATIC}
MICRO	inc.	micro	2.54GHz	789	261	502	1908	856	24437	8852	8701
MICRO	inc.	micro	2.4GHz	709	237	424	1647	763	22078	8816	7830
MICRO	inc.	micro	2.26GHz	621	208	357	1463	693	19525	8580	7036
MICRO	inc.	micro	2.13GHz	566	193	357	1334	620	17676	8553	6272
MICRO	inc.	micro	2.0GHz	506	174	253	1108	542	15911	8572	5590
MICRO	inc.	micro	1.86GHz	433	148	261	995	476	13770	8232	4896
MICRO	inc.	micro	1.6GHz	336	113	184	769	360	10613	8680	3508
MICRO	inc.	micro	1.2GHz	239	80	148	582	266	7848	10268	2581
MICRO	inc.	micro	1.06GHz	208	72	131	500	238	6896	10273	2176
MICRO	inc.	micro	0.93GHz	177	60	103	404	198	5944	10270	1792
MICRO	inc.	micro	0.8GHz	148	55	106	362	169	5069	10815	1309
MICRO-LIN	linear	micro	2.54GHz	1044	-373	-305	528	89	20795	7965	9674
FE	linear	spec	2.54GHz	623	-	-	-	-	-	-	10444
+FSB	linear	spec	2.54GHz	1318	-	-	-	-	-	9725	9430
ALL	linear	spec	2.54GHz	-621	1325	2809	3416	3750	36983	16892	6845

where the P_i of each component is the same of the single core model but the formulas to calculate the AR_{ij} should be modified to perform per core accounting. This is straightforward since PMCs already support per core event masks. Besides, it is needed to redefine the P_{static} component of the model because that component represents the static power of the entire processor. As a result, the P_{static} value obtained from single-core model training set is not valid because such value accounts for shared resources that consume static power that should not be replicated –i.e. the `L2` cache–. To recalculate it, we need an input data where all the cores modeled in the architecture are active. For that purpose, we re-executed the `MIX` subset of the microbenchmark suite on both cores at the same time in order to get a suitable training data. Then, a new P_{static} was generated by using a linear regression and dividing the obtained value by the number of cores.

3. VALIDATION AND EVALUATION

The model evaluation is organized in three main sections. The first one covers the `MICRO` model validation for one core, analysing their accuracy and responsiveness compared to other models. The second part validates the `MICRO` model for the whole chip, when two cores are active. And the third part validates the `MICRO` models for all the frequencies studied.

3.1 One Core Validation

3.1.1 Model Error

In this section we compare the `FE`, `+FSB`, `ALL`, `MICRO-LIN` and `MICRO` models in terms of accuracy (average error). Table 4 shows the average error and standard deviation for every application and every model. In general, all models behave similarly, showing average errors below 3%. However, their standard deviation differs showing that the more components tracked, the lesser the deviation is. This remarks the importance of taking into account as many power components as possible.

One interesting observation is how the error evolves taking into account that there is an incremental relation between the `FE`, the `+FSB` and the `ALL` and `MICRO-LIN` models. Depending on the modeled application, not all components affect the power modeling in the same manner. For the case of `+FSB`, several applications are modeled more accurately, showing high error reductions. Examples of this behavior are the `435.gromacs`, `453.povray`, `470.lbm` and `471.omnetpp` benchmarks. But other applications like the `434.zeusmp` and `437.leslie3d` do not expose any improvement, showing higher error rates for the `+FSB` model. This suggests that not every application is equally conditioned by the components in the architecture. This supports the idea of having a per component based model in order to never discard any architectural component that has a significant power contribution for a particular application. This is the case of the `ALL` and `MICRO-LIN` which in overall present lower error rates.

Table 4 also includes the average error for every model. Notice that for all of them the average error is always below a 3%. Initially, these results suggest that accuracy is not significantly improved by the component decomposition. However, the deviation is reduced when more components are taken into account recommending the usage of as much components as possible. Besides, there is one qualitative difference between them: the `MICRO` model can account for the power contribution of every component and the rest of the models cannot. In any case, if one just validates the models only using prediction accuracy, all models are valid and expose a similar error levels. Next section proves this assumption to be false on the basis of power phase detection and analysis.

3.1.2 Model Responsiveness

As we stated earlier, the *responsiveness* of a model is its capacity to react in a similar fashion as power, having similar inflection points with similar magnitudes. Therefore, to evaluate the responsiveness of a model, we apply a phase detection algorithm to both, the modeled and the real power. Then, we can compare them afterwards. The phase detection algorithm used is a modified version of the first pivot clustering algorithm presented in [15]. In our method, the power of a new sample is compared to the power of the current pivot sample, the starter of a phase. If the power of the new sample is within a specified threshold distance of the pivot power, in our case ±250mW, it is assigned to that phase. Otherwise, a new phase is added with the current sample as the pivot. We choose a ±250mW threshold because it only represents 2.24% of the average power consumption of the SPECcpu2006 suite. Lowering that threshold would introduce inaccuracies because it will be below the 2% error of the measurement device. The characteristics of each phase are its starting time (timestamp of the pivot sample), its duration, its variation with respect to the

Table 4: Average error of the single core `FE`, `FSB`, `ALL`, `MICRO-LIN` and `MICRO` models for the SPECcpu2006 suite.

Benchmark	Model FE		Model +FSB		Model ALL		Model MICRO-LIN		Model MICRO	
	%err	σ	%err	σ	%err	σ	%err	σ	%err	σ
400.perlbench	3.31	1.26	4.09	1.56	1.12	2.66	2.78	1.7	1.48	1.77
401.bzip2	1.2	1.61	1.22	1.7	3.47	2.91	1.21	1.65	1.26	1.59
403.gcc	2.93	3.25	3.69	3.93	3.52	5.16	3.65	4.24	3.4	3.52
410.bwaves	6.07	1.54	8.24	2.58	4.01	1.79	7	2.15	4.89	1.62
416.gamess	0.98	0.84	1.1	1.04	1.11	1.21	0.93	1.13	0.88	1.03
429.mcf	1	1.66	1.59	2.82	1.15	3.98	4.19	2.97	1.28	2.6
433.milc	1.28	1.09	2.48	1.91	1.25	2.25	1.51	1.67	1.21	1.28
434.zeusmp	1.5	1.96	3.47	3.53	2.4	3.03	2.21	2.74	1.64	2.21
435.gromacs	12.12	1.54	7.7	1.96	5.96	2.69	8.39	1.88	5.69	2
436.cactusADM	4.14	2.02	3.56	2.16	3.91	3.59	4.16	2.14	2.18	1.98
437.leslie3d	0.56	0.53	3.4	2.48	1.79	1.92	1.51	1.84	1.14	1.1
444.namd	1.34	1.07	1.91	1.23	2.39	2.19	2.08	1.32	2.11	1.39
445.gobmk	3.49	1.05	5.37	1.64	1.32	3.77	3.35	1.42	2.76	1.58
450.soplex	1.24	1.05	1.88	1.91	2.03	2.49	2.31	1.76	0.99	1.22
453.povray	0.6	0.36	0.42	0.65	0.56	0.48	0.34	0.6	0.32	0.53
454.calculix	0.5	2.54	4.85	2.59	3.24	2.7	2.75	1.89	0.87	2.71
456.hmmer	0.44	1.06	0.43	0.93	0.56	1.03	3.08	1.1	1.65	1.08
458.sjeng	3.33	0.78	5.04	0.98	0.5	1.32	4.73	0.93	3.34	0.92
459.GemsFDTD	6.32	1.48	8.03	2.05	1.33	1.6	7.07	1.77	2.56	1.78
462.libquantum	6.37	4.41	9.94	5	3	3.94	10	4.73	3.99	4.31
464.h264ref	3.14	1.03	5.13	1.58	6	1.8	1.79	1.32	3.47	1.38
465.tonto	1.2	2.28	1.48	2.57	2.31	4.25	1.49	2.55	1.41	2.52
470.lbm	18.95	0.82	3.95	1.4	2.85	1.99	5.8	1.37	6.87	1.48
471.omnetpp	4.74	0.75	2.7	0.9	2.8	1.42	2.6	0.78	3.6	0.82
473.astar	2.51	2.18	2.57	2.96	1.67	1.92	3.03	1.85	1.9	2.21
482.sphinx3	1.05	0.88	0.77	1.06	2.99	1.66	0.8	0.99	0.56	0.57
TOTAL	**2.15**	**4.11**	**2.77**	**2.57**	**2.02**	**1.48**	**2.65**	**2.45**	**1.89**	**1.64**

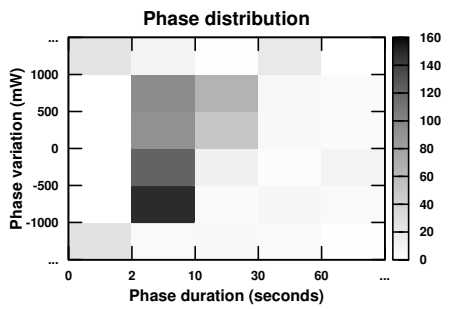


Figure 5: Power phases distribution of the SPECcpu2006.

prior power phase and the average power consumption for all the samples in the phase.

We classify the power phases in intervals of duration and variation. This allows us to perform analysis by duration and variation, while still being able to easily summarize the power trends. Figure 5 shows the distribution of the phases by its duration and its variation. From the duration point of view, we can observe that most of the phases last from 2 to 10 seconds. From the other point of view, we realize that most phases are defined after inflection points where power varies between 500mW and 1000mW. However, it is important to remark that power variations of more than 1000mW exist. These power variations correspond to significant targets of power aware policies. We observe that they are concentrated in the 2-or-less-seconds category, although they also appear for phases of 30-60 seconds. Next, we show that there are several phases which have a high variation and sufficient time to be detected and potentially treated by power aware policies.

Table 5 details the phase characterization for each benchmark sorted by number of phases. Columns 2 to 7 show the number of phases classified by the its variation. It also shows that most of the phases correspond to inflection points of more than 500mW of variation. The next 5 columns show the number of phases classified by its duration. We observe that most phases last from 2 to 10 seconds. The last column is the average power consumption. The `470.lbm` consumes the maximum power on average –about 14.5 watts– and `435.gromacs` the minimum, about 10.5 watts. The first one, shows high activity ratios in all the components and the second one has low activity ratios in the memory hierarchy and the BPU. At first sight, we see that half of the benchmarks are regular in power consumption, and that few ones concentrate the majority of inflection points. If we analyze the variability, we see that about a 14% of the inflections points have a variation bigger than 1 watt. The `454.calculix`, `403.gcc` and `410.bwaves` centralize the majority of such points with 44 (22+22), 19 (10+9) and 25 (7+18) respectively. The percentage of points with a variation within 0.5 and 1 watts is about 46% of the total. The benchmarks that contribute more to this category are `444.namd` (52+48), `434.zeus` (41+42), `462.libquantum` (19+25) and `465.tonto` (19+19). These benchmarks are also the ones that contribute more to the inflection points with a variation less than 0.5 watts, which account for the 40% of the total. Taking into account phase duration, we see that a 63% of phases are within the (2,10] range and that only a 6.9% of them are shorter than two seconds. This distribution shows that there is room to apply power aware policies since most of the phases are enough long to overcome the overhead of applying them. Moreover, we see that the set of aforementioned benchmarks are also the ones which experiment more shorter power phases. A special case is `454.calculix`, which shows an high number of very short and long phases. In particular, it presents 21 phases less or equal to 2 seconds and

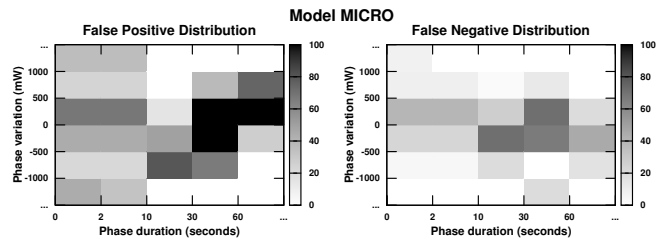
Table 5: Power phases characteristics of the SPECcpu2006.

	Phase variation(mW)						Phase duration (seconds)					
BENCHMARK	**< = -1000**	**(-1000,-500]**	**(-500,0]**	**(0,500]**	**(500,1000]**	**>1000**	**<=2**	**(2,10]**	**(10,30]**	**(30,60]**	**>60**	**POWER(mW)**
434.zeusmp	0	41	57	48	42	1	1	170	18	0	0	**11778.8**
444.namd	0	52	17	23	48	0	0	75	65	0	0	**11264.9**
454.calculix	22	5	4	4	6	22	21	17	0	25	0	**13157.1**
462.libquantum	2	19	19	24	25	2	1	83	6	1	0	**10636.5**
403.gcc	10	9	10	9	9	9	6	35	14	1	0	**11679**
410.bwaves	7	5	0	0	6	18	16	9	3	8	0	**11082.1**
465.tonto	2	19	13	19	19	0	2	55	11	4	0	**12131**
416.gamess	0	7	4	1	8	0	0	2	9	4	5	**12124.9**
429.mcf	2	4	2	2	0	3	2	2	2	5	2	**11453**
400.perlbench	1	1	7	6	1	2	2	9	7	0	0	**12628.4**
433.milc	0	3	1	0	3	1	1	3	0	1	3	**11665.7**
401.bzip2	0	1	5	6	3	0	0	4	5	1	5	**11888.2**
473.astar	0	1	2	3	1	0	0	0	1	2	4	**12413.9**
456.hmmer	0	1	2	2	1	0	0	3	2	0	1	**12425.9**
450.soplex	0	1	1	1	1	0	0	1	1	0	2	**11659.2**
436.cactusADM	0	0	1	2	0	0	0	3	0	0	0	**11507.2**
482.sphinx3	0	0	1	2	0	0	0	0	0	0	3	**11959.1**
459.GemsFDTD	0	0	1	1	0	0	0	1	0	0	1	**10999.4**
470.lbm	0	0	1	1	0	0	0	1	1	0	0	**14410.6**
471.omnetpp	0	0	1	0	0	0	0	0	0	0	1	**12445.8**
435.gromacs	0	0	0	0	0	0	0	0	0	0	0	**10573.8**
437.leslie3d	0	0	0	0	0	0	0	0	0	0	0	**10863.6**
445.gobmk	0	0	0	0	0	0	0	0	0	0	0	**12453.6**
453.povray	0	0	0	0	0	0	0	0	0	0	0	**12266.8**
458.sjeng	0	0	0	0	0	0	0	0	0	0	0	**12510.7**
464.h264ref	0	0	0	0	0	0	0	0	0	0	0	**12109.6**
TOTAL	**46**	**169**	**149**	**154**	**173**	**58**	**52**	**473**	**145**	**52**	**27**	

25 between half and one minute long. We have checked that its power consumption is periodic, switching between a very short phase and a long phase with a variation bigger than 1 watt.

After analyzing the insights of the `SPECcpu2006` power behavior, we use these results to validate that our power models predict a similar number of phases, with similar variations and durations. For this purpose we use two metrics. The main metric is the percentage of accuracy, which indicates the percentage of times that a model predicts correctly a real power phase. The other metric is the percentage of false positives (the model predicts an nonexistent change of phase). Again, the analysis are summarized taking into account the variation and the duration of the phases.

Figure 6 shows the percentage of accuracy for the model `FE`, model `+FSB` and model `MICRO`. The x and y axes correspond to the phase duration and the phase variation respectively. The z axis is the percentage of accuracy. The surface is colored to facilitate the readability. The darker the color, the more accurate the model is for that duration and variation. The three models show a similar trend. The bigger the variation, the bigger the accuracy is. These means that important changes in power consumption are detected even though all the components are not modeled. However, the final accuracy varies considerably among the different models. The `MICRO` model is almost 100% accurate for changes bigger than 1 watt. However, the `FE` models shows a 58% accuracy on average and the `+FSB` model about a 80%. This difference is important because our model is almost 100% accurate on detecting most important power variations where potentially power aware policies need to be reevaluated. If we take into account the rest of the phases we also see that the `MICRO` model is also the most accurate. In overall, the `MICRO` model shows a 83.31% of accuracy. The `FE` and the `+FSB` models show a 15.22% and a 27.9% of accuracy, respectively. This revokes the assumption that a power model with a low average error is valid for any applicability. We have shown that for phase detection analysis, a model should



Phase Duration	Phase variation (mW)											
	< = -1000		(-1000,-500]		(-500,0]		(0,500]		(500,1000]		>1000	
	FN	FP	FN	FP	FN	FP	FN	FP	FN	FP	FN	FP
<=2	0	12	0	0	0	0	0	0	0	0	2	9
(2,10]	0	2	7	33	28	54	36	62	9	23	0	4
(10,30]	0	0	1	4	10	7	14	8	2	0	0	0
(30,60]	1	0	0	6	2	11	5	10	1	3	0	0
>60	0	0	1	0	5	3	1	5	0	3	0	0

Figure 7: Distribution of False Positives and False Negatives for the model MICRO.

include as many components as possible.

If we analyze the other metric, the `MICRO` model shows a 34% of false positives and the `FE` and `+FSB` models show 10.81% and a 23.77% respectively. These last two models are less responsive, and therefore they are likely to not have false positives at the expenses of having plenty of false negatives.

We have also studied the distribution of false positives and negatives. Figure 7 shows it for the `MICRO` model. In the figure, the false positives are concentrated in the phases with a low variation or short duration, demonstrating that the false positive ratio is not an issue for power aware policies. Actually, the model performs fairly well for the phases and variations which are interesting for power policies. The rest of the models are not depicted due to space limitations but we point out some their trends. The `+FSB` model and the `FE` model show less false positives in overall, but have some of them in phases with huge variations. The reason is that sometimes there is a huge variation in the activity of the

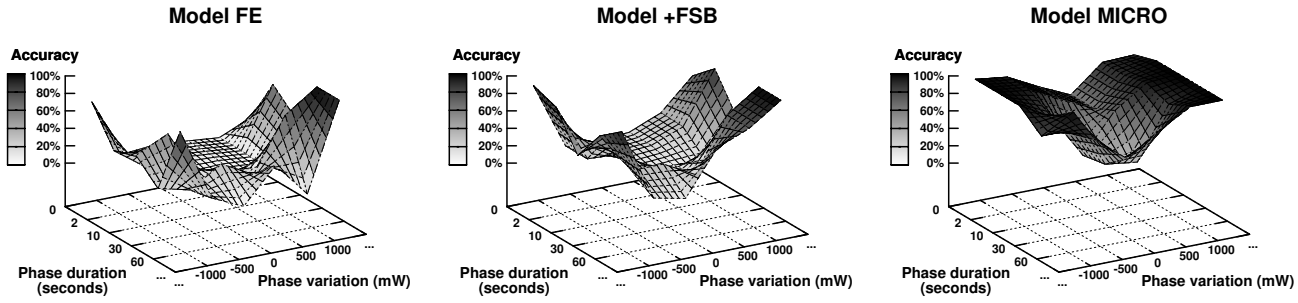


Figure 6: Accuracy distribution for three of the models studied.

components modeled, but the activity of the non-modeled components cancel (example 3 of Figure 4) the effect on power. As a result, such models predict non-existent huge changes.

3.2 Two Core Validation

We select a subset of the `SPECcpu2006` benchmarks to evaluate the power models when two cores are active and one application runs on each core. The applications are chosen according to the classification and characterization of the `SPECcpu2006` for multi-programming evaluation [26, 25]. We select a total number of 6 applications and for every pair of applications (15 experiments), we run each for ten minutes on one dedicated core.

Table 6 shows the power and model average errors on two cores. The models were extended as explained in Section 2.4.2. In general the `MICRO` model outperforms the other two and shows an average error of 4.63% while the other two models suffer from average errors of 5.12% and 6.09%, with higher standard deviations. This magnitudes are mainly caused by the pairs generated in conjunction with the `462.libquantum` application. For these cases, all models produce inaccurate estimations. We suspect that the reason is related to the hardware pre-fetcher because it affects heavily the application performance, compared to the rest of benchmarks. The rest of pairs generally expose an acceptable error (below the 5% border).

In conclusion, the extended `MICRO` model succeeds in modeling the entire multi-core chip. The last three columns of Table 6 show a direct application of the it: per core/per application power consumption. Notice that this model is the only one that can deliver this data.

3.3 Frequency validation

Table 7 summarizes the average error of the `MICRO` models. For all the frequencies, the `MICRO` models obtain low percentages of error, which decreases with frequency. The absolute error remains in the same magnitude for all the frequencies, showing that the methodology used is valid for any given power configuration.

To sum up, we have applied the proposed methodology on a particular architecture and we have validated that the power models generated are valid in terms of accuracy for different DVFS configurations. Moreover, a case study of responsiveness analysis was presented for a particular frequency, 2.53GHz. Overall, the models present fairly good results, outperforming the ones generated using previous modeling techniques.

Table 7: Average error summary of the `MICRO` models for all the frequencies.

Frequency (Ghz)	Avg.Power (mW)	Model MICRO %err	σ	err(mW)
0.800	2113.46	6.07	6.96	128.28
0.933	2654.6	4.76	5.22	126.42
1.066	3120.04	4.1	4.33	127.96
1.200	3584.31	3.56	3.73	127.61
1.600	4795.22	3.27	3.73	156.59
1.866	6407.61	2.47	2.59	158.03
2.000	7240.7	2.23	2.36	161.79
2.133	8100.77	2	2.34	161.85
2.266	9023.32	1.89	2.12	170.61
2.400	10006.21	1.9	1.96	189.64
2.533	11143.27	1.89	1.64	210.38
AVERAGE		**2.85**	**1.4**	**154.15**

4. RELATED WORK

In general, previous works on power modeling focus in two main aspects: power model generation and its usage to guide power aware policies. For instance, Tao Li et al. [19] study the power consumption at the OS level. For this purpose, an IPC based model is built based on the broadly accepted observation that there is a linear relation between IPC and power consumption. In [3], F. Bellosa implements an OS power aware policy based on data collected at runtime through the PMCs. In this work, overall power consumption is being modeled using a reduced set of PMCs. K.Singh et al. [31] describe a methodology based on a set of microbenchmarks that stress particular components of the processor architecture being modeled. As a result, a model suitable for any workload –generic– is produced. Our main difference with all these approaches is that they use the PMCs –one or several– to predict the power consumption of the whole processor, treating it as a black-box.

A. Miyoshi et al. [21] introduce the concept of critical power slope. This new metric is used to determine the effectiveness of activating power saving techniques on scheduling points. A contribution of this work is that depending on the workload characteristics lowering the processor frequency results in power savings. Our proposal agrees with this result, and points out the fact that power models have to be validated at inflection points of power in order to be used by power aware policies.

The works of R.Joseph, C.Isci, G.Contreras and M.Martonosi [16, 14, 9] present many similarities to our proposal because we follow the same objective: generate an accurate power model capable to breakdown the power consumption.

Table 6: Average errors for the two core modeling and accounting for the power contribution of each core.

Benchmark	Power	Model MICRO		Model FE		Model +FSB		Processor	Core 0	Core 1
cpu0–cpu1	(mW)	%err	σ	%err	σ	%err	σ	(mw)	(mw)	(mw)
400.perlbench-436.cactusADM	18540.20	4.10	1.85	4.16	1.38	3.56	1.41	19298.02	10707.80(55.49%)	8590.22(44.51%)
400.perlbench-453.povray	19686.18	6.08	1.09	0.52	0.53	0.60	0.58	20882.35	10762.56(51.54%)	10119.79(48.46%)
400.perlbench-454.calculix	20462.40	4.38	1.52	2.33	0.97	5.10	1.94	21352.66	10768.96(50.43%)	10583.71(49.57%)
400.perlbench-462.libquantum	24034.74	6.80	1.78	17.46	1.21	13.54	1.77	22398	10954.22(48.91%)	11443.79(51.09%)
400.perlbench-473.astar	19377.92	5.62	3.11	2.71	1.89	1.91	1.33	20444.1	10776.56(52.71%)	9667.53(47.29%)
462.libquantum-436.cactusADM	22661.80	9.22	2.08	13.78	1.54	8.96	2.12	20569.68	11432.35(55.58%)	9137.33(44.42%)
462.libquantum-453.povray	23896.88	9.02	1.38	16.37	1.25	13.34	1.55	21739.6	11508.96(52.94%)	10230.64(47.06%)
462.libquantum-454.calculix	24716.67	10.15	1.45	15.12	1.73	9.31	2.01	22206.72	11495.94(51.77%)	10710.77(48.23%)
462.libquantum-473.astar	23353.39	9.26	2.00	17.01	1.69	11.70	2.97	21188.89	11399.75(53.80%)	9789.13(46.20%)
473.astar-436.cactusADM	18964.17	4.40	3.76	3.01	1.80	3.04	2.15	18131.4	9523.93(52.53%)	8607.47(47.47%)
473.astar-453.povray	20049.61	3.12	2.87	3.21	3.35	3.62	3.40	19776.85	9619.77(48.64%)	10157.08(51.36%)
473.astar-454.calculix	20798.19	3.04	3.59	3.01	2.73	3.78	1.79	20206.38	9628.47(47.65%)	10577.91(52.35%)
436.cactusADM-453.povray	18477.38	1.47	1.09	5.09	1.50	4.36	1.50	18708.42	8590.34(45.92%)	10118.09(54.08%)
436.cactusADM-454.calculix	19269.42	1.68	1.17	6.06	2.41	8.81	3.08	19143.36	8569.32(44.76%)	10574.04(55.24%)
453.povray-454.calculix	20163.70	2.84	0.89	3.46	1.30	6.78	2.52	20732.2	10142.99(48.92%)	10589.20(51.08%)
AVERAGE		**4.63**	**2.9**	**5.12**	**6.31**	**6.09**	**4.15**			

The most similar work to ours is [14]. In that work, the Pentium IV architecture is decomposed in microarchitectural components in a similar fashion. But the model generation methodology differs: they use component area as the main heuristic to derive each component power consumption, as well as some microbenchmarks in order to guide the manual tuning of the model. In contrast, we derive the marginal effect of each component on power from data gathered using a specifically designed set of microbenchmarks. Thus, we avoid their test-and-evaluate manual tuning and the need of the floorplan information. In [9], they also use statistical linear regression to generate the model but they validate it against a similar set of applications. In overall, these works are able to predict power consumption in the same order of accuracy than the ones that use less PMCs but, they can breakdown the power consumption. These works also proved that the usage of sampling techniques to read a bigger set of PMCs does not incur in inaccuracies. In [15], C.Isci and M.Martonosi present a study comparing power phase classification techniques based on PMCs and control flow information, concluding that PMCs detect more power phases.

In [27], M.Powell proposes a methodology based on few performance counters to estimate the activity factors of several microarchitectural structures. Then, they estimate the power consumption of such structures based on the the activity estimations. However, they do not evaluate empirically their proposal because a simulation based infrastructure is required to be able to obtain such low level information.

There is not doubt about the applicability of PMC-based power models since they provide accurate insight into processor power consumption and that they can be used to breakdown the power consumption of a given platform. While this work is based on the same idea of using the PMCs for power estimation, there are some key characteristics that differentiate our proposal from all previous research:

- We present a new methodology to model current multicore architectures. The key difference is that our methodology is strictly based on the appropriate definition of power components and the specifically designed microbenchmarks that isolate and decouple each component activity. This allows us to produce models that are accurate, responsive and decomposable.
- Besides the validation of the model based on average error, we *validate and evaluate the responsiveness of the model*, comparing it against other PMC-based power models.
- We present a wide set of power models for the *Intel® Core^{TM} 2 processor*, a high performance processor designed with power efficiency as a main design constraint, for different DVFS configurations.

5. CONCLUSION

In this paper we presented a novel systematic power modeling methodology based on performance monitoring counters (PMCs) to derive per component power breakdowns on current multicore architectures. Moreover, besides validating the accuracy of power estimates, we validate the models by evaluating their *responsiveness*, their capacity to detect power variations. We generated a set of models for the Intel® Core^{TM} 2 Duo architecture, modeling one or two cores for different DVFS configurations. The models were validated using empirical measurements of long executions of the `SPECcpu2006` benchmarks, and despite the energy-efficient characteristics and the complexity of the modeled architecture, the models shown fairly good accuracy and responsiveness. Concretely, the accuracy validation showed average errors between [1.89-6]%, and the responsiveness validation when operating at maximum frequency showed an overall 83.31% accuracy on phase change predictions and almost 100% accuracy for changes bigger than 0.5 watts. As a result, we showed that the model produced is sensible to variations as small as 0.25 watts, which only represent a 2.24% variation of the average power consumption of the `SPECcpu2006` suite. Moreover, the comparison performed against models built using existing methods showed that our methodology produces more accurate and responsive power models. In overall, this work addresses the problem of the lack of decomposable power models for current architectures.

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